Laser-Based SLAM Using GMapping in ROS for Indoor Environment Mapping

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***Abstract*—Simultaneous Localization and Mapping (SLAM) is a fundamental capability required for autonomous mobile robots operating in unknown environments. In this project, a laser-based SLAM system is implemented using the GMapping algorithm within the Robot Operating System (ROS). An indoor dataset containing laser scan and odometry information is used to generate a two-dimensional occupancy grid map. The SLAM pipeline is evaluated using front-facing laser scan data, and the resulting map quality is analyzed qualitatively. The results demonstrate that GMapping is capable of producing a coherent indoor map using limited sensor information, while also revealing common SLAM challenges such as pose drift and incomplete coverage.**

***Index Terms*—SLAM, GMapping, ROS, LiDAR, Occupancy Grid Mapping**

I. Introduction

Autonomous mobile robots must be capable of navigating and understanding unknown environments without prior maps. Simultaneous Localization and Mapping (SLAM) addresses this challenge by enabling a robot to estimate its own pose while simultaneously constructing a map of the surrounding environment. Laser-based SLAM methods are particularly effective in indoor environments due to the accuracy and reliability of LiDAR sensors.

This project focuses on implementing a two-dimensional SLAM system using the GMapping algorithm within the Robot Operating System (ROS). The primary objective is to generate an occupancy grid map of an indoor environment using laser scan and odometry data and to analyze the performance of the SLAM system through qualitative evaluation.

II. SLAM Algorithm Details

GMapping is a two-dimensional SLAM algorithm based on Rao-Blackwellized Particle Filters (RBPF). In this approach, each particle represents a possible robot trajectory along with an associated map estimate. Odometry data is used as the motion model to propagate particles, while laser scan measurements are used to update particle weights through scan matching.

At each time step, particles are evaluated based on how well the incoming laser scan aligns with the existing map. Particles with higher likelihoods are retained through resampling, while low-probability particles are discarded. The final map is generated from the highest-weight particle, resulting in a probabilistic occupancy grid representation of the environment.

III. Dataset Description

The dataset used in this project consists of recorded ROS bag files containing laser scan and odometry data collected in an indoor environment. The dataset includes front-facing laser scan data published on the `/front_scan` topic and odometry information published on the `/odom` topic.

Although rear-facing laser scan data is available, the primary experiments presented in this report utilize only the front-facing laser data. This choice allows for a focused evaluation of SLAM performance using a single sensor configuration.

IV. Graphical User Interface

RViz is used as the primary graphical user interface for visualizing the SLAM process. The occupancy grid map generated by GMapping is displayed using the `/map` topic, while laser scan data and odometry are visualized using the `/front_scan` and `/odom` topics, respectively. RViz provides real-time visualization of map formation, robot pose estimation, and sensor alignment.

V. Implementation Details

This section describes the complete implementation workflow used to perform 2D SLAM using the ROS framework and the `gmapping` algorithm. The implementation is based on offline sensor data replay, simulation-time synchronization, and visualization using RViz.

A. Dataset Preparation and ROS Bag Generation

The provided dataset consisted of multiple CSV files containing time-stamped sensor data, including odometry, front laser scan, rear laser scan, and IMU measurements. Since the SLAM algorithm in ROS operates on ROS topics, the raw CSV files were converted into a ROS-compatible bag file.

A Python script, `convert_csv_to_bag.py`, was executed to parse the CSV files and publish the data to appropriate ROS topics such as `/odom`, `/front_scan`, and `/rear_scan`. During this process, a static transform between the robot base frame and the laser sensor frames was also generated. The published topics were recorded into a single ROS bag file named `output.bag`. Terminal output confirmed successful conversion with 100% completion for odometry, front scan, and rear scan data streams.

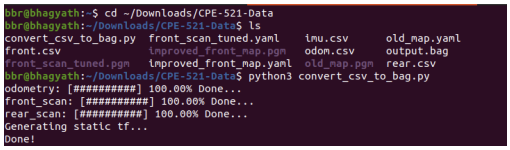

Fig. 1. ROS bag file generation from CSV sensor data

B. ROS Environment Initialization

The ROS environment was initialized by starting the ROS master using the `roscore` command. Since the SLAM process relied on prerecorded data rather than live sensor input, ROS was configured to operate in simulation time by setting the parameter `use_sim_time` to true. This ensured that all ROS nodes synchronized with the clock published from the bag file during playback.

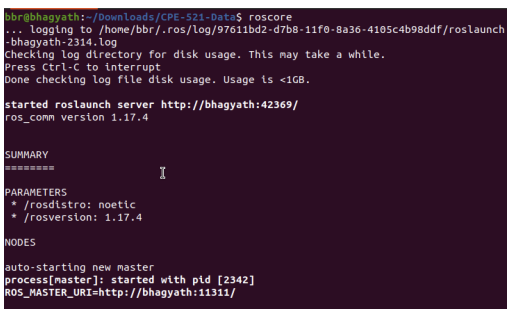

Fig. 2. Initialization of the ROS master using the `roscore` command

C. Bag File Playback

The generated ROS bag file (`output.bag`) was replayed using the `rosbag play --clock` command. This command published the sensor and odometry data at their original recorded timestamps while also broadcasting the simulation clock. Terminal logs confirmed successful opening of the bag file and continuous publishing of messages during playback.

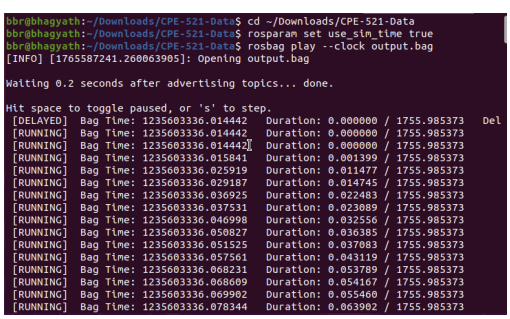

Fig. 3. ROS bag playback with simulation clock enabled

D. SLAM Execution Using gmapping

The SLAM algorithm was implemented using the `slam_gmapping` node from the `gmapping` package. The algorithm was configured to use the front-facing laser scan topic (`/front_scan`) as input. Upon execution, gmapping initialized the particle filter, registered the first laser scan, and continuously updated the robot pose and occupancy grid map as new scans were received.

Terminal messages indicated successful scan registration, pose updates, and scan matching scores, demonstrating stable SLAM operation throughout the playback duration. The front laser scan was selected for the final implementation due to its better environmental coverage and more consistent scan overlap compared to the rear laser scan.

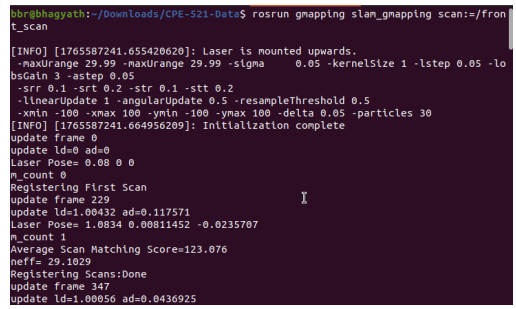

Fig. 4. GMapping SLAM execution using front laser scan

E. Visualization Using RViz

RViz was used as the graphical user interface to visualize the SLAM process. The fixed reference frame was set to `map` to ensure consistent visualization across all data sources. The following display elements were enabled:

- **LaserScan**: Subscribed to the `/front_scan` topic and displayed using flat square markers with intensity-based coloring.
- **Map**: Subscribed to the `/map` topic to visualize the occupancy grid generated by gmapping, with a resolution of 0.05 meters per cell.
- **Odometry**: Subscribed to the `/odom` topic and visualized as an arrow representing the robot pose and trajectory.
- **Axes and Grid**: Enabled to provide spatial reference and orientation context.

RViz status indicators confirmed that all subscribed topics were successfully received and correctly transformed into the map frame.

Fig. 5. RViz startup and initialization

Although the dataset contained both front and rear laser scan data, the SLAM implementation in this work was intentionally restricted to the front-facing laser scan (`/front_scan`).

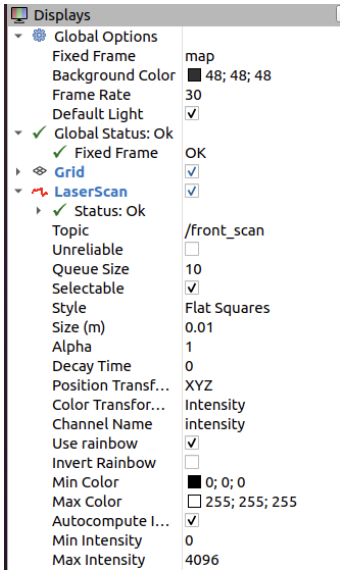

Fig. 6. Front laser scan visualization in RViz

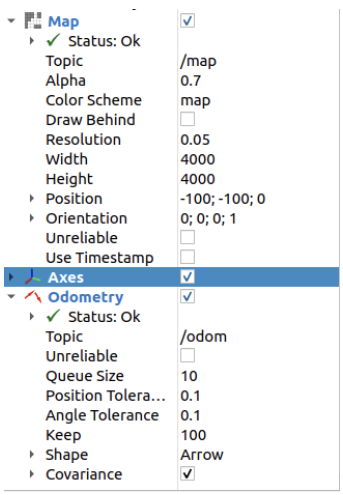

Fig. 7. Occupancy grid map and odometry visualization in RViz

The front laser provided consistent forward-facing coverage, stronger overlap between consecutive scans, and more reliable geometric features for scan matching.

Preliminary visualization of the front laser data in RViz confirmed stable alignment with the robot odometry frame and continuous environmental coverage. In contrast, using a single, well-aligned laser source simplified the SLAM pipeline and reduced frame transformation ambiguities.

Therefore, all mapping results presented in this report are generated exclusively using the front laser scan data, ensuring a stable and reproducible SLAM configuration.

F. Map Output and Documentation

The final output of the SLAM process was an occupancy grid map published on the `/map` topic. As the bag playback progressed, the map incrementally expanded and converged to a stable representation of the environment. Screenshots were captured at key stages of execution, including ROS initialization, bag playback, SLAM execution, and RViz visualization. These screenshots are referenced in the Simulation Results section to support qualitative performance analysis.

VI. Simulation Results and Performance Discussion

A. Baseline Front Laser Scan Mapping

The SLAM process was first executed using a baseline configuration in RViz with default GMAPPING parameters and only the front-facing laser scan topic (`/front_scan`). This initial setup served as a reference to evaluate the raw mapping performance without parameter tuning.

The resulting occupancy grid map captures the overall structure of the indoor environment, including corridors, intersections, and room boundaries. Major walls are clearly visible and generally well aligned, indicating successful scan matching and pose estimation during normal forward motion.

However, several limitations are observed in the baseline map. Wall edges appear slightly blurred in certain regions, particularly near corners and during sharp robot rotations. Cone-shaped distortions are noticeable in areas with rapid angular motion, suggesting reduced scan overlap between consecutive laser measurements. Additionally, small pose drift accumulates along long corridors due to odometry uncertainty. Since only a front-facing laser scanner was used, areas outside the sensor's field of view remain unmapped, resulting in incomplete environmental coverage.

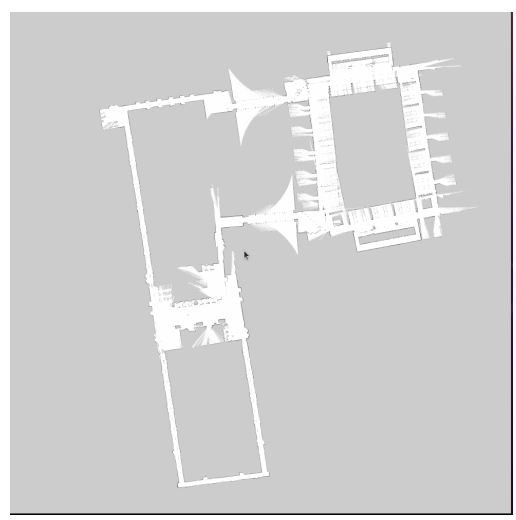

Fig. 8. Baseline front laser scan mapping result

B. Tuned Front Laser Scan Mapping

After evaluating the baseline SLAM results, the GMAPPING configuration was refined to improve map clarity and consistency. The tuning process focused on modifying the scan integration frequency while keeping the overall SLAM pipeline unchanged. Specifically, the `linearUpdate` parameter was set to 0.5 m, allowing the algorithm to update the map

after shorter translational motion of the robot. This adjustment increases the overlap between consecutive laser scans and enables more frequent pose corrections.

All other parameters, including angular update thresholds, map resolution, and particle count, were kept at their default values to ensure a fair comparison between the baseline and tuned configurations. The front-facing laser scan topic (`/front_scan`) continued to be used as the sole sensor input.

With the tuned configuration, the resulting occupancy grid map exhibits noticeable improvements compared to the baseline case. Wall boundaries appear sharper and more continuous, particularly along straight corridors and at intersections. Distortions observed during sharp rotations in the baseline configuration are reduced, indicating improved scan alignment and more stable pose estimation.

Although minor artifacts remain due to odometry drift and the limited field of view of the front-facing laser scanner, the tuned results demonstrate more consistent mapping performance overall. This improvement confirms that adjusting the `linearUpdate` parameter plays a critical role in enhancing SLAM performance when operating with a single laser sensor.

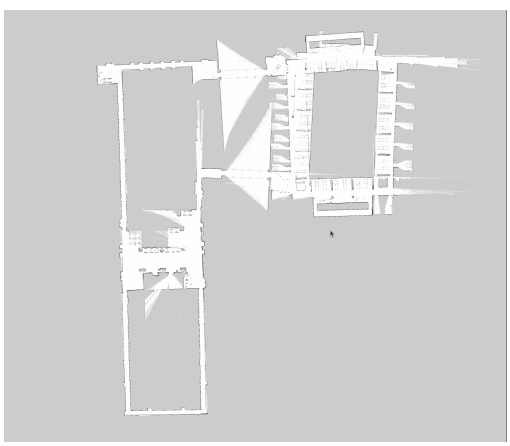

Fig. 9. Tuned front laser scan mapping result

C. Qualitative Analysis of Mapping Performance

A qualitative comparison was performed to evaluate the mapping performance of the baseline and tuned GMAPPING configurations using the front-facing laser scanner. The analysis focuses on visual map characteristics, including wall sharpness, structural consistency, continuity of corridors, and the presence of mapping artifacts.

In the baseline configuration, the generated map captures the overall layout of the environment; however, wall boundaries appear slightly blurred in certain regions. Wedge-shaped artifacts are visible during sharp robot rotations, and small discontinuities are observed along long corridors. These effects indicate limited scan overlap and accumulated odometry error between consecutive scan updates.

After tuning the `linearUpdate` parameter to 0.5 m, the qualitative map quality improves noticeably. Wall boundaries become more clearly defined, and corridor structures exhibit better continuity. The frequency of wedge-shaped distortions is reduced, suggesting improved scan alignment during rotational motion. The tuned configuration also shows reduced pose drift over extended trajectories, resulting in a more coherent overall map.

Despite these improvements, minor artifacts remain due to the inherent limitations of using a single front-facing laser scanner. Regions outside the sensor's field of view remain unmapped, and small residual distortions persist in areas with aggressive motion. Nevertheless, the tuned configuration provides a more stable and visually consistent representation of the environment, demonstrating the effectiveness of parameter tuning for enhancing SLAM performance under limited sensor conditions.

VII. Discussion and Lessons Learned

This project highlights the critical role of sensor configuration, data quality, and parameter tuning in achieving reliable SLAM performance. Although the GMAPPING algorithm is capable of generating coherent occupancy grid maps using limited sensor input, the completeness and accuracy of the resulting maps are inherently constrained by the field of view of the laser scanner. Using only a front-facing laser limits environmental coverage and introduces challenges in areas that are not directly observed by the sensor.

The results demonstrate that careful parameter tuning can significantly enhance map quality even without modifying the underlying SLAM algorithm. In particular, adjusting the scan update frequency improved scan overlap, reduced mapping artifacts, and enhanced wall continuity. This emphasizes the importance of algorithm configuration in practical SLAM deployments, especially when operating with constrained sensor setups.

Through this project, valuable hands-on experience was gained in implementing SLAM within the ROS framework, managing prerecorded sensor data, and diagnosing mapping artifacts arising from odometry drift and sensor limitations. Beyond ideal simulation scenarios, the work provided practical insight into real-world SLAM challenges, reinforcing the need for informed parameter selection and sensor-aware system design. Future improvements could include incorporating additional sensors or multi-directional laser data to further improve mapping robustness and environmental coverage.

VIII. Conclusion

This project presented the implementation and evaluation of a 2D SLAM system using the GMapping algorithm within the ROS framework. Using prerecorded odometry and laser scan data, an occupancy grid map of an indoor environment was successfully generated and analyzed.

The SLAM process was initially executed using a baseline configuration with a front-facing laser scanner and later evaluated qualitatively to assess mapping accuracy and limitations.

The resulting maps captured the overall structure of the environment, including corridors and room boundaries, while revealing artifacts such as pose drift and incomplete coverage due to the limited sensor field of view.

Overall, this work demonstrates the effectiveness of GMapping for indoor mapping using single-direction laser data while highlighting the importance of sensor configuration and parameter selection. Future improvements could include the integration of additional sensors, loop-closure methods, or alternative SLAM algorithms to further enhance map accuracy and robustness.

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